Assignment 3- Networking

SUBMITTED BY :- AASTHA

USER ID :- 68493

BATCH ID :- 25S

BEGINNER FRIENDLY

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Why IPv6 Is Important

Every device connected to the internet — your phone, laptop, smart TV, even your fridge — needs a unique address. This is called an **IP address**. The older system, **IPv4**, uses 32-bit addresses and can support about **4.3 billion** unique addresses. That sounded like a lot in the 1980s, but today we have tens of billions of connected devices.

IPv6 is the newer system that uses 128-bit addresses. **IPv6 (Internet Protocol version 6)** is the latest version of the Internet Protocol used to identify devices and allow them to communicate over the internet. It is important because the older version, IPv4, has several limitations.

📦 Huge Address Space (Main Reason)

- IPv4 uses **32-bit addresses** → about **4.3 billion addresses**
- IPv6 uses **128-bit addresses** → about **340 undecillion addresses (very huge!)**

🚀Supports IoT (Internet of Things)

- Modern devices like smart watches, smart homes, sensors need unique IPs
- IPv6 allows **every device to have its own unique address**

🔒 Better Security

IPv6 was designed with built-in security features, making connections safer by default.

IPv6 has **built-in security features** like IPSec (Internet Protocol Security)

Think of IPv4 as a small town with limited house numbers. IPv6 is like an entire planet where every grain of sand gets its own address.

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What Is the Purpose of a Subnet Mask?

A **subnet mask** is like a postal code for your network. It tells your device which part of an IP address identifies the **network** and which part identifies the specific **device** on that network.

A **subnet mask** is a 32-bit number used in IPv4 that separates an IP address into:

- **Network part** (identifies the network)
- **Host part** (identifies the device inside that network)

👉 Example:

- IP Address: **192.168.1.10**
- Subnet Mask: **255.255.255.0**

1. Identify Network and Host

The main purpose is to divide an IP address into:

- Network ID
- Host ID

👉 Why important:

- Helps devices understand **which network they belong to**

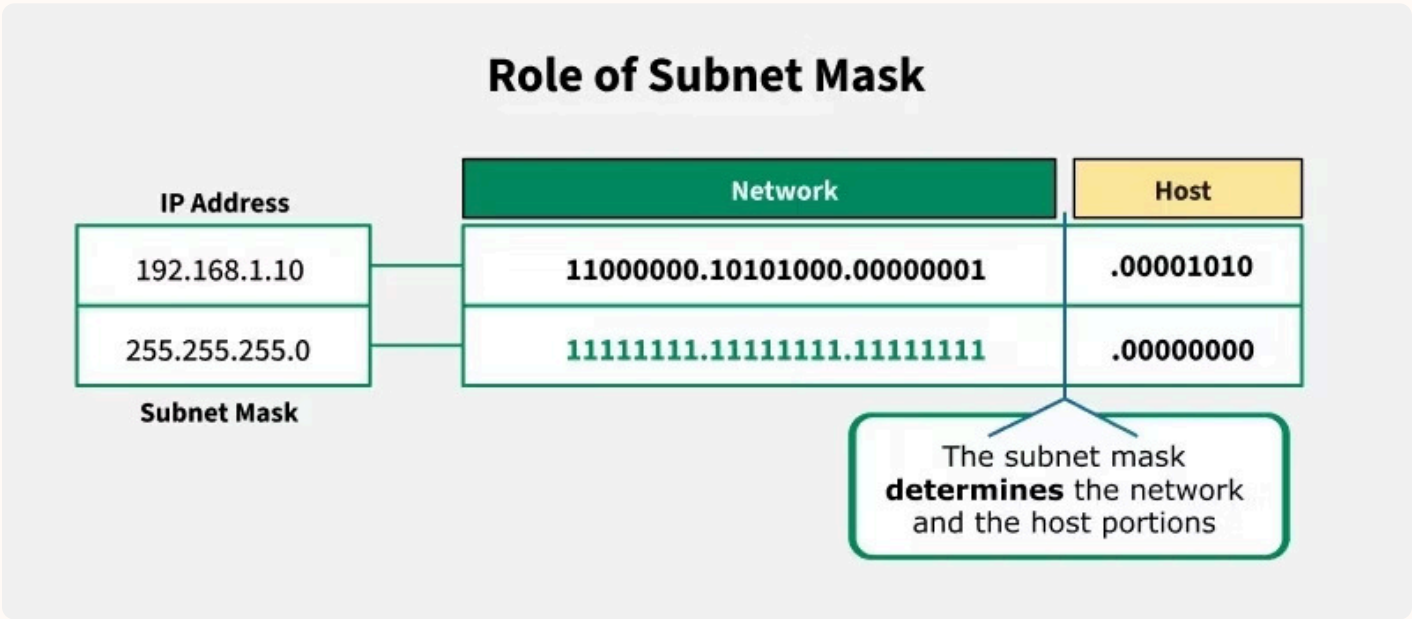
2. Helps in Communication

Subnet mask helps a device decide:

- Send data **directly** (same network)
- Send data via **router** (different network)

4. Improves Network Security

- Different subnets can be separated
- Access can be controlled between them



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How NAT Helps in Home Networks

NAT (Network Address Translation) is the reason all your devices can share a single internet connection. It is a technique used in home networks to allow multiple devices to share a single public IP address when accessing the internet.

1. Saves Public IP Addresses

- One internet connection → many devices
- Efficient use of limited IPv4 addresses

2. Allows Multiple Devices to Use Internet

👉 Example:

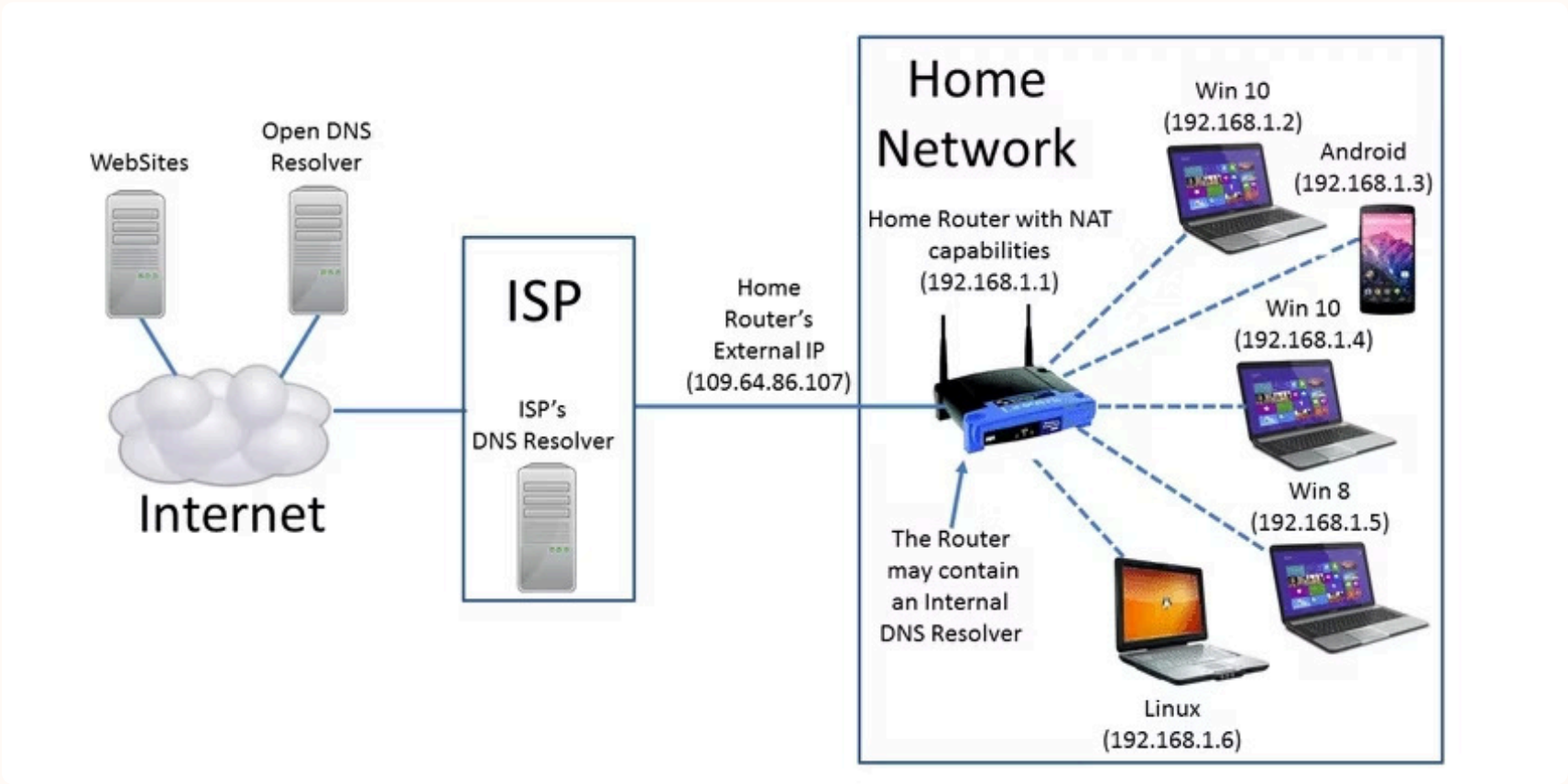
- Mobile, laptop, smart TV, CCTV, tablet
  All can access internet simultaneously

3. Provides Basic Security

- Internal IPs are **hidden from the internet**

👉 Benefit:

- Hackers cannot directly access your devices



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